

Ashwat N

AI Engineer | Machine Learning | Generative AI

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Profile

Computer Science undergraduate and AI/ML developer with industry experience in LLMs, VLMs, NLP, RAG, and backend development.

Professional Experience

AI/Software Intern, TNQ Tech

01/2026 – 09/2026
Coimbatore, India

- Worked with **LLMs, VLMs, and OCR systems** for AI-based data processing.
- Developed **VLM-based image understanding and structured data extraction pipelines** for multimodal workflows.
- Built AI-powered software workflows involving **LLM processing, OCR, and audio/transcription** for confidential projects.

Education

B.E. in Computer Science Engineering, KGISL Institute of Technology

Tamil Nadu, India

Skills

Programming Python, C, SQL	Backend & Databases FastAPI, SQLAlchemy, PostgreSQL, MySQL, FAISS, ChromaDB	ML/AI NLP, Computer Vision, OCR, PyTorch, TensorFlow, Scikit-learn
Cloud & Tools Firebase, Git, VS Code	Generative AI LLMs, VLMs, RAG, Agentic AI, MCP, ADK	

Achievements

Hackathon
Winner of Hackathon conducted by Kongu Engineering College (KEC)

OCI Generative AI Course & OCI AI Foundation Course
Completed OCI AI Foundations and Generative AI training in AI/ML, LLMs, RAG, and OCI AI services.

Projects

Hokorobi  *AI-Powered Developer Problem Discovery*

- Built an NLP pipeline using **transformers, embeddings, and RAG** to identify and structure technical problems from online discussions.
- Used **MiniLM embeddings and FLAN-T5** to transform unstructured discussions into structured, developer-ready problem statements.

AI-Driven Behaviour Analysis System for Crime Interrogation 

- Built real-time facial expression analysis using **DeepFace and OpenCV** with continuous emotion tracking.
- Integrated **Google Gemini** for behavioural analysis and Flask/Firebase for case management and automated reporting.

AI Powered Plasma Reactor 

- Built a linear surrogate model mapping **4 tokamak coil-current inputs to 8 plasma observables**, achieving **R² > 0.87 across 7/8 variables**.
- Integrated the surrogate with an **RL-ready control interface** for plasma-control experiments using Python, NumPy, Pandas, Scikit-learn and TORAX.